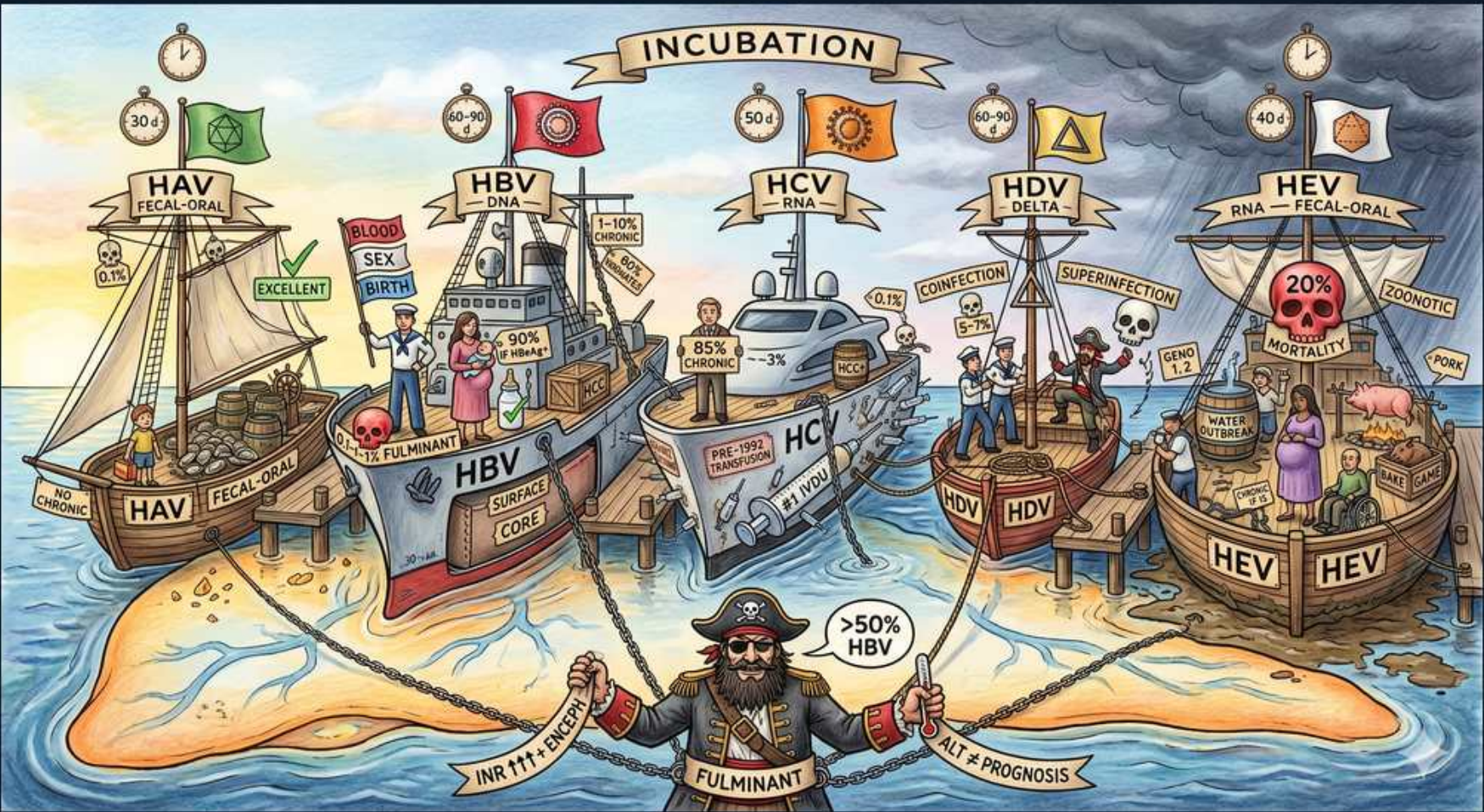


Viral Hepatitis — The 5 Types (A,B,C,D,E)



1. HAV ship — fecal-oral

WHAT YOU SEE

Leftmost ship labeled HAV with FECAL-ORAL banner

WHAT IT ENCODES

Hepatitis A is a non-enveloped picornavirus (positive-sense RNA) spread fecal-oral via contaminated food/water, shellfish, and during travel to endemic areas. Incubation is ~2-6 weeks (shorter than HBV). It causes an acute, self-limited illness; viremia and fecal shedding peak BEFORE jaundice, so patients are most contagious in the pre-icteric prodrome. Diagnosis = anti-HAV IgM (acute); anti-HAV IgG = past infection or vaccination.

BOARD TRAP

HAV never becomes chronic. A question describing a returning traveler with 'chronic hepatitis A and progression to cirrhosis' is wrong — HAV has no carrier state. Pick a different virus (HBV/HCV) for any chronicity stem.

2. HAV good prognosis

WHAT YOU SEE

Green EXCELLENT / checkmark flag on HAV ship

WHAT IT ENCODES

HAV recovers fully with lifelong immunity; there is no carrier state and no cirrhosis. Fulminant hepatitis is rare (<1%) and rises with age >50 or pre-existing liver disease (e.g., underlying HCV). A cholestatic or relapsing course can occur over weeks-months but still resolves.

BOARD TRAP

Don't choose 'liver transplant for chronic HAV cirrhosis' — HAV does not chronify. The only HAV scenario needing transplant is rare acute fulminant failure, not chronic disease.

3. HBV ship — DNA virus

WHAT YOU SEE

Second ship labeled HBV with DNA banner and surface/core barrels

WHAT IT ENCODES

Hepatitis B is the ONLY DNA hepatitis virus — a partially double-stranded hepadnavirus that replicates via reverse transcriptase (so nucleos(t)ide analogs like tenofovir/entecavir work). It carries surface antigen (HBsAg, the envelope) and core antigen (HBcAg, internal); HBeAg is a secreted marker of replication. Incubation is long (~1-4 months).

BOARD TRAP

'All hepatitis viruses are RNA' is the distractor — HCV, HDV, HAV, and HEV are RNA; only HBV is DNA. Also remember HBsAg-negative does NOT exclude infection during the window period.

4. HBV transmission — blood/sex/birth

WHAT YOU SEE

BLOOD, SEX, BIRTH labels on HBV ship cargo/figures

WHAT IT ENCODES

HBV spreads parenterally (needlestick, IVDU, transfusion), sexually, and perinatally (vertical) — NOT fecal-oral. It is far more infectious than HIV or HCV per needlestick (~30% for HBeAg+ source vs ~1.8% HCV, ~0.3% HIV). Household/sexual contacts and infants of HBsAg+ mothers are the key exposed groups for prophylaxis.

BOARD TRAP

Calling HBV 'fecal-oral' is wrong — that's HAV and HEV. Also, sexual transmission is much more prominent for HBV than for HCV; don't swap those two.

5. HBV chronicity by age

WHAT YOU SEE

'90%' marker near infant/birth figure on HBV ship

WHAT IT ENCODES

Risk of chronic HBV is inversely age-dependent: ~90% if acquired perinatally/neonatally, ~25-50% in early childhood, but only ~5% if acquired as an immunocompetent adult. This is because neonatal immune tolerance prevents clearance. Hence the urgency of HBIG + vaccine at birth for infants of HBsAg+ mothers.

BOARD TRAP

Don't quote the 90% chronicity figure for an adult-acquired infection — adults clear >95% of acute HBV. The 90% number applies to neonatal/perinatal transmission only.

6. HCV yacht — 85% chronic

WHAT YOU SEE

Central yacht labeled HCV with '85% CHRONIC' placard

WHAT IT ENCODES

Hepatitis C is a flavivirus (RNA) that becomes chronic in ~75-85% of those infected — the most chronicity-prone hepatitis virus and the leading cause of chronic viral hepatitis, cirrhosis, and liver transplant in the West. Acute infection is usually asymptomatic, which is why one-time screening of all adults 18-79 is recommended. Cure with direct-acting antivirals (e.g., sofosbuvir-based regimens) achieves SVR >95%.

BOARD TRAP

HCV — not HBV — is the leading cause of chronic viral hepatitis and HCC in the United States/West. Don't pick HBV for 'most common cause of chronic hepatitis in the West.'

7. HCV blood-borne

WHAT YOU SEE

Needle/transfusion cues on HCV yacht

WHAT IT ENCODES

HCV is primarily blood-borne — IVDU is now the dominant route, with pre-1992 transfusion as the classic historical cohort. Sexual transmission (~low, rises with HIV co-infection/MSM) and perinatal transmission (~5%) are uncommon compared with HBV. Extrahepatic associations: mixed cryoglobulinemia, membranoproliferative glomerulonephritis (MPGN), porphyria cutanea tarda.

BOARD TRAP

Acute HCV rarely causes fulminant liver failure — its danger is chronic progression, not ALF. If a stem has acute fulminant hepatitis, think HAV/HBV/HEV or acetaminophen, not HCV alone.

8. HDV — coinfection

WHAT YOU SEE

HDV ship with COINFECTION placard (left figure)

WHAT IT ENCODES

HDV (delta agent) is a defective circular RNA virus that **REQUIRES** the HBV surface antigen (HBsAg) to assemble and infect — it cannot replicate alone. COINFECTION = simultaneous acquisition of HBV + HDV; this raises fulminant risk acutely but, because clearing HBV clears HDV, it is usually self-limited with low chronicity.

BOARD TRAP

HDV cannot infect an HBsAg-negative person — it needs the HBV envelope. A 'delta hepatitis in a non-HBV patient' stem is impossible. Coinfection (acquired together) is the **LESS** chronic one; superinfection is the dangerous chronic form.

9. HDV — superinfection

WHAT YOU SEE

SUPERINFECTION placard on right of HDV ship

WHAT IT ENCODES

HDV SUPERINFECTION = delta acquired by someone who is already a chronic HBV carrier. Because HBsAg is already abundant, HDV replicates aggressively → high rate of chronic HDV (~80%), accelerated cirrhosis, and increased fulminant/ALF risk. Suspect superinfection when a stable chronic HBV patient suddenly worsens. Diagnose with anti-HDV/HDV RNA.

BOARD TRAP

Distinguish superinfection (chronic HBV carrier + new HDV = dangerous, chronic) from coinfection (both at once = mostly self-limited). A chronic HBV patient with abrupt clinical deterioration = HDV superinfection, not a new HAV.

10. HEV ship — fecal-oral

WHAT YOU SEE

Rightmost ship HEV with FECAL-ORAL/RNA banner

WHAT IT ENCODES

Hepatitis E is an RNA virus (hepevirus) transmitted fecal-oral via contaminated water (epidemic genotypes 1/2 in Asia/Africa) and via undercooked pork/game (zoonotic genotypes 3/4 in developed countries). Like HAV it is usually acute and self-limited with no chronic state in immunocompetent hosts.

BOARD TRAP

HEV is fecal-oral like HAV, **NOT** blood-borne. Exception to remember: HEV **CAN** become chronic in solid-organ transplant/immunosuppressed patients — treat with ribavirin and reduce immunosuppression.

11. HEV 20% pregnancy mortality

WHAT YOU SEE

Red skull '20% MORTALITY' near pregnant figure on HEV ship

WHAT IT ENCODES

The signature HEV board fact: fulminant hepatitis with ~15-25% mortality in pregnant women, especially the 3rd trimester. A pregnant traveler returning from an endemic area with acute hepatitis and high mortality risk = HEV until proven otherwise.

BOARD TRAP

The HEV danger is acute fulminant disease in PREGNANCY, not chronicity. Don't attribute high pregnancy mortality to HAV — it's HEV specifically.

12. Fulminant pirate — INR/enceph

WHAT YOU SEE

Central pirate holding chains with FULMINANT and INR↑↑↑ + ENCEPH banners

WHAT IT ENCODES

Acute liver failure (fulminant hepatitis) = hepatic encephalopathy **PLUS** coagulopathy (INR ≥1.5) developing within weeks in a patient without pre-existing liver disease. Among viral causes, HBV (especially with HDV coinfection/superinfection) drives the majority; HEV in pregnancy is another key cause; HCV almost never. Management = ICU + urgent liver transplant evaluation.

BOARD TRAP

'HCV is the top cause of fulminant viral hepatitis' is wrong — HBV (often with HDV) leads, and HCV rarely causes ALF. Remember encephalopathy is **REQUIRED** to label it ALF; coagulopathy alone is not enough.